

Workshop Practice Lab:

1. A workshop practice lab is a laboratory designed to facilitate the application of theoretical knowledge to practical tasks.
2. It is typically used in engineering and other technical fields, to provide students with hands-on experience in their chosen field.
3. The lab typically consists of various tools, such as saws, drills, lathes, and other machinery, as well as safety equipment, such as goggles and protective clothing.
4. The lab also contains a variety of materials, such as wood, metal, plastic, and other substances, to enable students to practice their skills.
5. The lab is supervised by a qualified instructor, who provides guidance and instruction to the students.
6. The instructor may also provide demonstrations, lectures, and other activities to help the students understand the concepts and techniques they are learning.
7. The lab is designed to provide a safe and controlled environment for students to practice their skills, while minimizing the risk of injury or damage to equipment.
8. Workshop practice labs are an important part of any engineering or technical program, and are essential for students to gain the skills and knowledge they need to succeed in their chosen field.

S. No. Mechanical Workshop Duration

1. Mechanical workshops can range in duration from a single day to several weeks, depending on the complexity of the project.
2. A typical mechanical workshop will cover topics such as basic mechanical engineering principles, mechanical design, materials selection, and manufacturing processes.
3. The duration of a workshop will depend on the number of topics to be covered, the complexity of the topics, and the level of detail required.
4. In a one-day workshop, topics may include an overview of mechanical engineering principles, basic design principles, and an introduction to materials selection and manufacturing processes.
5. In a multi-day workshop, topics may include more in-depth discussions of mechanical engineering principles, more advanced design principles, and detailed discussions of materials selection and manufacturing processes.
6. In a multi-week workshop, topics may include more detailed discussions of mechanical engineering principles, advanced design principles, and detailed discussions of materials selection, manufacturing processes, and system integration.

1 Introduction to Mechanical Workshop Material, Tools and Machines 3 Hrs

- Mechanical workshops are areas where mechanical engineering is practiced.
- They contain a variety of tools and machines used to create and modify mechanical parts, components, and assemblies.
- Common tools and machines found in a mechanical workshop include lathes, milling machines, grinders, drill presses, and saws.
- Lathes are used to shape and cut metal components.
- Milling machines are used to shape and cut metal components.
- Grinders are used to sharpen and refine metal components.
- Drill presses are used to drill holes in metal components.
- Saws are used to cut metal components.
- Other tools used in a mechanical workshop include calipers, micrometers, and gauges.
- Calipers are used to measure the size of components.
- Micrometers are used to measure the thickness of components.
- Gauges are used to measure the pressure of components.

To study layout, safety measures and different engineering materials used in workshop

1. Layout:
 - A workshop layout is an important part of any workshop, as it helps ensure that the space is used efficiently and that safety is maintained.
 - The layout should take into account the size and shape of the workshop, the type of work to be done, and any equipment that will be used.
 - The layout should also be designed to allow for easy access to tools and materials.
2. Safety Measures:
 - All workshops should have appropriate safety measures in place, including appropriate lighting, ventilation, and fire safety equipment.
 - Protective clothing and equipment, such as safety glasses, gloves, and protective footwear, should also be worn.
 - A first aid kit and other emergency equipment should also be available.
3. Engineering Materials:
 - Mild Steel: Mild steel is a low-carbon steel and is one of the most common materials used in workshops. It is easy to work with and is relatively inexpensive.
 - Medium Carbon Steel: Medium carbon steel is a higher-carbon steel and is stronger than mild steel. It is more difficult to work with and is more expensive than mild steel.
 - High Carbon Steel: High carbon steel is the strongest of the three

types of steel and is the most difficult to work with. It is also the most expensive.

- High Speed Steel: High speed steel is a type of steel that is designed to be used at high speeds. It is more expensive than other types of steel.
- Cast Iron: Cast iron is a type of iron that is melted and poured into a mold to form a shape. It is strong and durable, but is difficult to work with.

To study and use of different types of tools, equipment, devices & machines used in fitting, sheet metal and welding section

- Fitting: Tools such as hammers, pliers, wrenches, screwdrivers, and chisels are commonly used in fitting.
- Sheet Metal: Sheet metal tools such as shears, snips, notchers, and presses are used to cut, shape, and form sheet metal.
- Welding: Welding equipment such as welding torches, welding rods, and safety equipment are essential for welding.

To determine the least count of Vernier calliper, vernier height gauge, micrometer (Screw gauge) and take different reading over given metallic pieces using these instruments

- Vernier Caliper: It is a measuring instrument which is used to measure the internal and external dimensions of an object. It consists of two scales, the main scale and the vernier scale. The main scale is graduated in centimeters (or millimeters) and the vernier scale is graduated in tenths of a millimeter or hundredths of a centimeter. The least count of vernier caliper is 0.1 mm or 0.01 cm.
- Vernier Height Gauge: It is a measuring instrument used to measure the height of an object. It consists of a main scale and a vernier scale. The main scale is graduated in centimeters and the vernier scale is graduated in tenths of a millimeter or hundredths of a centimeter. The least count of vernier height gauge is 0.1 mm or 0.01 cm.
- Micrometer (Screw Gauge): It is a measuring instrument which is used to measure the diameter of an object. It consists of a main scale and a vernier scale. The main scale is graduated in millimeters and the vernier scale is graduated in thousandths of a millimeter. The least count of micrometer (screw gauge) is 0.001 mm.

To take different readings over given metallic pieces using these instruments, the object should be placed in between the jaws of the instrument and the reading should be taken by noting the main scale and the vernier scale. The reading should be noted in the form of the main scale reading plus the vernier scale reading. For example, if the main scale reading is 5 cm and the vernier scale reading is 0.4 cm, then the reading should be noted as 5.4 cm.

2 Machine Shop 3 Hrs

1. A machine shop is a facility that houses a variety of machines and tools used for metalworking and other related activities.
2. The machines found in a machine shop typically include lathes, milling machines, drill presses, and grinders.
3. A machine shop can also contain other equipment such as welding machines, saws, and sanders.
4. In order to maintain the machines and tools in a machine shop, it is important to keep them clean and lubricated.
5. The machines should be inspected regularly to ensure that they are functioning properly.
6. It is also important to use the correct safety equipment when operating the machines in a machine shop.
7. The three main types of machine shops are manual, CNC, and robotic.
8. Manual machine shops are the most basic type of machine shop and are used for simple tasks such as drilling and tapping.
9. CNC machine shops are more advanced and can be used for more complex tasks such as milling and turning.
10. Robotic machine shops are the most advanced type of machine shop and are used for tasks such as welding and assembly.
11. In order to become proficient in operating the machines in a machine shop, it is important to have a good understanding of the different types of machines and their functions.
12. It is also important to have a good understanding of the safety protocols and procedures that are necessary when operating the machines.
13. With proper training and experience, a person can become an expert in operating the machines in a machine shop.

Demonstration of Working, Construction and Accessories for Lathe Machine

A lathe machine is a machine tool used for shaping and machining metal, wood, and other materials. It is commonly used for facing, plane turning, step turning, taper turning, threading, knurling, and parting.

- **Construction of Lathe Machine:** The main components of a lathe machine are the bed, headstock, tailstock, carriage, and tool post. The bed is the foundation of the machine and supports all other components. The headstock contains the spindle, which rotates the workpiece. The tailstock provides support to the workpiece and can be adjusted to different positions. The carriage holds the cutting tool and is moved along the bed to complete the machining operation. The tool post holds the cutting tools.
- **Working of Lathe Machine:** The workpiece is placed between the headstock and tailstock and is secured in place. The cutting tool is mounted on the tool post and is moved along the bed by the carriage. The spindle

is rotated at a certain speed and the cutting tool is fed into the workpiece to shape it.

- **Accessories for Lathe Machine:** A number of accessories are available for lathe machines, such as live centers, faceplates, steady rests, and follow rests. Live centers are used to support the workpiece in the headstock and tailstock. Faceplates are used to hold the workpiece in place. Steady rests and follow rests are used to support the workpiece during machining.

3 Fitting Shops in 3 Hours

1. **The Tailor's Workshop:** Located in the heart of the city, The Tailor's Workshop offers custom-made clothing and tailoring services. They specialize in suits, blazers, and other formal attire. They also provide alterations and repairs.
2. **The Seamstress:** The Seamstress is a small boutique offering a wide selection of ready-to-wear clothing, as well as custom-made clothing. They specialize in bridal wear and evening gowns.
3. **The Fabric Store:** The Fabric Store is a one-stop shop for all your fabric needs. They offer a wide selection of fabrics, patterns, and notions, as well as custom-made clothing and alterations. They also offer classes on sewing and pattern-making.

1. Practice marking operations

- Marking operations are the actions taken by a business to ensure that the quality of their products or services meets the standards set by the company.
- Quality assurance is an important part of any business, and it is important to ensure that all marking operations are done correctly.
- Common marking operations include inspection, testing, and sorting.
- Inspection involves looking at the product or service to ensure that it meets the required standards.
- Testing involves using a variety of methods to ensure that the product or service meets the required standards.
- Sorting involves separating the products or services into different categories based on quality.
- Quality assurance is a process, and it is important to regularly review the marking operations to ensure that they are effective.

2. Preparation of U or V -Shape Male Female Work piece

1. **Filing:** Filing is the process of removing material from a workpiece to create a desired shape or finish. It involves using a file, which is a metal tool with a series of small teeth, to remove material from the surface of the workpiece.

2. Sawing: Sawing is the process of cutting material into two or more pieces using a saw. It can be used to cut a variety of materials, including wood, metal, plastic, and stone.
3. Drilling: Drilling is the process of creating a hole in a workpiece by using a drill bit. The drill bit is a rotating tool that is used to cut into the material.
4. Grinding: Grinding is the process of removing material from a workpiece using an abrasive wheel. The abrasive wheel is a rotating wheel that is used to cut away material from the surface of the workpiece.

4 Carpentry Shop 3 Hrs

1. Carpentry is a craft that involves the construction of wooden structures and objects. It requires a high level of skill and precision to achieve the desired results.
2. Carpenters use a variety of tools to complete their work, such as saws, hammers, chisels, planes, and measuring tools. They also use a variety of materials, such as wood, metal, and plastic.
3. Carpenters must be familiar with the properties of different types of wood and how to work with them. They must also understand the principles of joinery, which is the process of connecting two pieces of wood together.
4. Carpenters must be able to read and interpret blueprints. They must also have knowledge of the building codes and regulations that apply to the work they are doing.
5. Carpentry can be a dangerous job, as carpenters must use sharp tools and work in awkward positions. Safety is of utmost importance and carpenters must always wear the proper safety equipment and follow safety protocols.
6. Carpenters typically work in a workshop or on a construction site. They often work in teams and must be able to communicate and collaborate effectively.
7. Carpenters typically work full-time and may be required to work overtime or on weekends. The average carpentry job requires 3 hours of work per day.

Study of Carpentry Tools, Equipment and different joints. Making of Cross Half lap joint, Half lap Dovetail joint and Mortise Tenon Joint

- Carpentry is a craft that involves the use of tools to construct items such as furniture, cabinets, and other objects.
- To complete carpentry projects, a carpenter must have the right tools, equipment, and knowledge of different joints.

- Cross half lap joints are used to join two pieces of wood together. To make this joint, the carpenter cuts a half-lap on each piece of wood and then fits them together.
- Half lap dovetail joints are used to join two pieces of wood together at a right angle. To make this joint, the carpenter cuts a half-lap on the end of each piece of wood and then fits them together.
- Mortise and tenon joints are used to join two pieces of wood together. To make this joint, the carpenter cuts a mortise into one piece of wood and a tenon into the other. The tenon is then inserted into the mortise and secured with glue.

5 Welding Shop 6 Hrs

1. Welding is a fabrication process that involves the use of heat to join metal components together.
2. It is a highly skilled craft that requires specialized tools and equipment.
3. The most common welding processes are arc welding, gas welding, and oxy-fuel welding.
4. In order to become a certified welder, one must pass a welding certification test and complete 6 hours of welding shop time.
5. Welding shop time is supervised by a qualified welding instructor and involves the use of welding tools, safety equipment, and welding techniques.
6. During the welding shop time, the student will learn about the different types of welding processes, welding terminology, and welding safety practices.
7. The student will also practice welding on various types of metal, such as aluminum, stainless steel, and mild steel.
8. Upon completion of the welding shop time, the student will receive a certification that is valid for two years.

Introduction to BI Standards and Reading of Welding Drawings

1. The International Institute of Welding (IIW) is a global professional organization that develops and promotes standards for welding and related technologies.
2. The IIW provides guidance on welding processes and procedures, including the use of welding symbols to communicate welding instructions.
3. Welding drawings provide a clear and concise description of the welding process, and the symbols used to represent the welding operations.
4. To understand welding drawings, it is important to be familiar with the basic symbols and terms used in welding, such as butt joint, lap joint, TIG welding, and MIG welding.
5. Butt joints are used to join two pieces of metal together, while lap joints are used to join two pieces of metal that overlap each other.
6. TIG welding is a welding process that uses a tungsten electrode to create an electric arc between the electrode and the metal being welded.

7. MIG welding is a welding process that uses a consumable wire electrode to create an electric arc between the electrode and the metal being welded.
8. Both TIG and MIG welding require the use of proper safety gear and procedures to ensure the safety of the welder.

6 Moulding and Casting Shop 6 Hrs

1. Moulding and casting is a process used to create objects from a variety of materials. It involves pouring molten material into a mould and allowing it to cool and harden.
2. The mould is usually made of metal or plastic and is designed to create a particular shape. It is important to use the right mould for the job, as the shape of the finished product will be determined by the shape of the mould.
3. The casting process involves heating the material to a high temperature before it is poured into the mould. This is usually done in a furnace or kiln.
4. After the material has been heated, it is poured into the mould and allowed to cool and harden. Depending on the type of material, this process can take anywhere from a few minutes to several hours.
5. Once the material has cooled and hardened, it is removed from the mould and is ready to be used. It is important to use the right tools and techniques when removing the material from the mould, as this can affect the quality of the finished product.
6. Moulding and casting is used in a variety of industries, including automotive, aerospace, and medical. It is also used to create sculptures, jewellery, and other decorative items.

Introduction to Patterns, pattern allowances, ingredients of moulding sand and melting furnaces. Foundry tools and their purposes Demo of mould preparation and Aluminum casting Practice – Study and Preparation of mould for Plastic

- Patterns are the physical representation of the object that needs to be cast. Pattern allowances are the extra material added to the pattern to account for shrinkage of the metal during the cooling process.
- The ingredients of moulding sand are silica, clay, and water. The sand is mixed and blended to create a homogeneous mixture. The sand is then packed into the mould to form a cavity.
- Melting furnaces are used to melt the metal and pour it into the mould cavity. Foundry tools are used to shape the molten metal and to remove the casting from the mould.
- The demo of mould preparation involves the use of a pattern, core boxes, flask, and moulding sand. The pattern is placed into the core box and the

sand is packed around it. The flask is then placed over the core box and the sand is packed into the flask.

- Aluminum casting is the process of pouring molten aluminum into a mould and allowing it to cool and solidify. The practice of pouring molten aluminum requires the use of safety equipment and proper technique.
- The study and preparation of mould for plastic involves the use of a pattern, core box, flask, and plastic resin. The pattern is placed into the core box and the plastic resin is poured into the flask. The flask is then placed over the core box and the plastic resin is poured into the mould. The mould is then allowed to cool and the plastic part is removed.

7 CNC Shop 6 Hrs

1. CNC stands for Computer Numerical Control, and is a type of manufacturing process that uses computer-controlled machines to fabricate parts from a variety of materials.
2. CNC machines are used in a wide range of industries, from aerospace to automotive, and are used to make parts for everything from medical implants to furniture.
3. CNC machines are operated by a computer program that is used to control the machine's movements and operations. The program is typically written in a language called G-code.
4. CNC machines require a skilled operator to ensure that the machine is working correctly and that the finished parts meet the required specifications.
5. CNC machines can be used to make parts from a variety of materials, including metals, plastics, and composites.
6. CNC machines are used in a variety of manufacturing processes, including milling, drilling, and turning.
7. CNC machines are typically operated in a shop environment, with a shop typically running for 6 hours a day.

Study of main features and working parts of CNC machine and accessories that can be used. Perform different operations on metal components using any CNC machines

1. CNC machines are computer-controlled machines that are used to perform different operations on metal components.
2. The main features of CNC machines include:
 - Programmable memory: This allows the machine to store and recall instructions and parameters.
 - Tool changer: This allows the machine to switch between different tools quickly and efficiently.
 - Feed rate control: This allows the machine to control the speed at which it moves the workpiece.

- Spindle speed control: This allows the machine to control the speed at which the cutting tool rotates.
- 3. CNC machines can be used to perform a variety of operations, including drilling, milling, tapping, and reaming.
- 4. Accessories that can be used with CNC machines include:
 - Tool holders: These are used to securely hold the cutting tool in place.
 - Workholding devices: These are used to securely hold the workpiece in place.
 - Coolant systems: These are used to keep the cutting tool and workpiece cool during operations.
 - Dust collectors: These are used to collect dust and debris generated during operations.
- 5. Safety is an important consideration when using CNC machines. It is important to wear appropriate safety equipment, such as safety glasses and hearing protection, when operating the machine.

8 To Prepare a Product Using 3D Printing 3 Hrs

1. 3D printing is a process of making three dimensional solid objects from a digital file.
2. 3D printing is achieved using additive processes, where successive layers of material are laid down in different shapes.
3. The most common 3D printing material is plastic, although other materials such as metal, ceramic and composite materials can also be used.
4. 3D printing is used in many industries for prototyping and production, including automotive, aerospace, medical, dental and consumer products.
5. In order to prepare a product using 3D printing, a 3D model must first be created. This can be done using computer aided design (CAD) software or 3D scanning.
6. Once the 3D model is created, it must be prepared for printing. This involves slicing the model into layers and generating a G-code file, which contains instructions for the 3D printer.
7. The G-code file is then sent to the 3D printer, which reads the instructions and prints the model layer by layer.
8. Depending on the size and complexity of the model, 3D printing can take anywhere from a few minutes to several hours.

Total 33 Hrs

1. Total 33 Hrs is a term used to refer to the total number of hours a person needs to spend in order to complete a task or project.
2. This number is calculated by taking into account the amount of time needed to complete the task, the amount of time needed to complete any preparatory work, and the amount of time needed for any additional activities such as research, planning, and review.

3. The total number of hours needed to complete a task or project can vary depending on the complexity and scope of the task.
4. In order to determine the total number of hours needed to complete a task or project, it is important to consider the amount of time needed to complete each individual step, as well as the amount of time needed to complete the entire task or project.
5. It is also important to consider any potential delays or interruptions that may occur during the completion of the task or project.
6. In addition, it is important to consider any potential changes or modifications that may need to be made during the completion of the task or project.
7. Once the total number of hours needed to complete a task or project is determined, it is important to plan ahead and make sure that the task or project is completed within the allotted time frame.
8. This will help to ensure that the task or project is completed on time and to the highest quality.

Course Outcome:

1. Understand the fundamental principles and concepts of computer science.
2. Develop the ability to design, implement, and evaluate computer-based solutions to problems.
3. Gain an understanding of the ethical and social implications of computer technology and its applications.
4. Acquire the skills necessary to effectively communicate with computer professionals and non-computer professionals.
5. Develop the ability to work collaboratively on teams.
6. Develop the capacity to think critically and creatively, and to reason logically.
7. Understand the impact of computer technology on society, and be able to use it responsibly.

The Students Will Be Able to:

- Use various engineering materials, tools, machines and measuring equipments (Blooms Level CO1)
- Perform machine operations in lathe and CNC machine (K3 CO2)
- Perform manufacturing operations on components in fitting and carpentry shop (K3 CO3)
- Perform operations in welding, moulding, casting and gas cutting (K3 CO4)
- Fabricate a job by 3D printing manufacturing technique K3 (K3 CO5)

Reference Books:

- **Organic Chemistry** by John McMurry
- **Inorganic Chemistry** by Gary L. Miessler and Donald A. Tarr

- **Physical Chemistry** by Atkins and de Paula
- **Analytical Chemistry** by Gary D. Christian
- **Biochemistry** by Jeremy M. Berg, John L. Tymoczko, and Lubert Stryer
- **Quantum Chemistry** by Ira N. Levine
- **Environmental Chemistry** by Stanley E. Manahan

1. Workshop Practice, H S Bawa, McGraw Hill

1. Workshop Practice provides a comprehensive introduction to the basics of workshop technology.
2. It covers topics such as tools and machines used in a workshop, materials used for manufacturing, and the principles of workshop design.
3. It also covers topics such as safety and health considerations, quality control, and the use of computers in the workshop.
4. The book is written by H S Bawa and published by McGraw Hill.
5. It is an essential resource for students and professionals in the field of workshop technology.

2. Mechanical Workshop Practice, K C John, PHI

1. Introduction to Mechanical Workshop Practice:
 - Workshop practice is an important part of engineering and technology education.
 - It is the practice of applying the principles of engineering, science and mathematics in the design and manufacture of machines and tools.
 - Workshop practice helps students to develop their skills in the design, construction, maintenance and operation of machines and tools.
2. Types of Workshop Practice:
 - Machine shop practice: This involves the use of power tools and machine tools to fabricate and assemble components.
 - Foundry practice: This involves the use of furnaces and casting techniques to produce castings from metals and alloys.
 - Sheet metal practice: This involves the use of tools and techniques to form and shape sheet metal components.
 - Welding practice: This involves the use of welding techniques to join metals and other materials.
3. Safety in the Workshop:
 - Safety is of paramount importance in the workshop.
 - All safety regulations and guidelines must be followed at all times.
 - The use of protective clothing, such as safety glasses, gloves, and aprons, is essential.
 - All tools and machines must be used in accordance with the manufacturer's instructions.
4. Workshop Tools and Equipment:
 - The workshop is equipped with a variety of tools and machines, such as lathes, milling machines, drill presses, grinders, and saws.

- Hand tools, such as chisels, hammers, files, and wrenches, are also used.
- The selection of tools and machines depends on the type of work to be done.

3. Workshop Practice Vol 1, and Vol 2, by HazraChoudhary, Media Promoters and Publications

1. Workshop Practice Vol 1 provides an in-depth look into the process of setting up and running a successful workshop. It covers topics such as:
 - Identifying and researching potential workshop topics
 - Creating an effective workshop agenda
 - Planning and preparing for the workshop
 - Running the workshop
 - Evaluating the workshop
2. Workshop Practice Vol 2 builds on the knowledge from Vol 1 and dives deeper into the topics of:
 - Developing a workshop team
 - Creating an effective workshop environment
 - Managing the workshop
 - Developing a post-workshop strategy
3. Both volumes of Workshop Practice provide valuable insights into the process of running a successful workshop and provide practical tips and techniques to help ensure the success of any workshop.

4. CNC Fundamentals and Programming

1. CNC stands for Computer Numerical Control. It is a technology that uses computer-controlled machines to perform operations such as cutting, drilling, and milling.
2. CNC programming involves writing instructions for a CNC machine to execute. The instructions are written in a programming language such as G-code, which is a standard language for CNC machines.
3. CNC machines can be used for a variety of operations, including drilling, milling, turning, and grinding.
4. CNC machines are used in a wide range of industries, from automotive to aerospace.
5. CNC programming involves writing instructions for a CNC machine to execute. The instructions are written in a programming language such as G-code, which is a standard language for CNC machines.
6. CNC machines can be programmed to produce parts with high accuracy and repeatability.
7. CNC machines require regular maintenance to ensure they are running optimally.
8. Safety is an important consideration when working with CNC machines. Proper safety procedures must be followed to ensure the safety of the

operator and other personnel.